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21 Authors

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22 Date of Revision

16 February 2009.

Alitame

1 Nonproprietary Names

None adopted.

2 Synonyms

Aclame; L-aspartyl-D-alanine-N-(2,2,4,4-tetramethylthietan-3-yl)amide; 3-(L-aspartyl-D-alaninamido)-2,2,4,4-tetramethylthietane.

3 Chemical Name and CAS Registry Number

L- α -Aspartyl-N-(2,2,4,4-tetramethyl-3-thietanyl)-D-alaninamide anhydrous [80863-62-3]

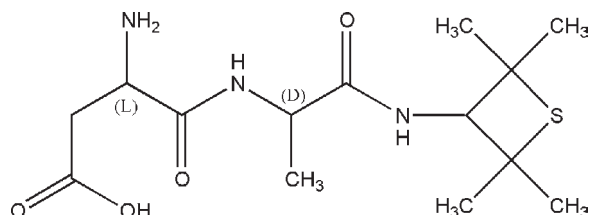
L- α -Aspartyl-N-(2,2,4,4-tetramethyl-3-thietanyl)-D-alaninamide hydrate [99016-42-9]

4 Empirical Formula and Molecular Weight

C₁₄H₂₅N₃O₄S 331.44 (for anhydrous)

C₁₄H₂₅N₃O₄S·2½H₂O 376.50 (for hydrate)

5 Structural Formula



6 Functional Category

Sweetening agent.

7 Applications in Pharmaceutical Formulation or Technology

Alitame is an intense sweetening agent developed in the early 1980s and is approximately 2000 times sweeter than sucrose. It has an insignificant energy contribution of 6 kJ (1.4 kcal) per gram of alitame.

Alitame is currently primarily used in a wide range of foods and beverages at a maximum level of 40–300 mg/kg.⁽¹⁾

8 Description

Alitame is a white nonhygroscopic crystalline powder; odorless or having a slight characteristic odor.

9 Pharmacopeial Specifications

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10 Typical Properties

Acidity/alkalinity pH = 5–6 (5% w/v aqueous solution)

Isoelectric point pH 5.6

Melting point 136–147°C

Solubility see Table I.

Table I: Solubility of alitame.

Solvent	Solubility at 20°C unless otherwise stated
Chloroform	1 in 5000 at 25°C
Ethanol	1 in 1.6 at 25°C
<i>n</i> -Heptane	Practically insoluble
Methanol	1 in 2.4 at 25°C
Propylene glycol	1 in 1.9 at 25°C
Water	1 in 8.3 at 5°C
	1 in 7.6 at 25°C
	1 in 3.3 at 40°C
	1 in 2.0 at 50°C

Specific rotation $[\alpha]_D^{25} = +40^\circ$ to $+50^\circ$ (1% w/v aqueous solution)

11 Stability and Storage Conditions

Alitame is stable in dry, room temperature conditions but undergoes degradation at elevated temperatures or when in solution at low pH. Alitame can degrade in a one-stage process to aspartic acid and alanine amide (under harsh conditions) or in a slow two-stage process by first degrading to its β -aspartic isomer and then to aspartic acid and alanine amide. At pH 5–8, alitame solutions at 23°C have a half-life of approximately 4 years. At pH 2 and 23°C the half-life is 1 year.

Alitame should be stored in a well-closed container in a cool, dry place.

12 Incompatibilities

Alitame may be incompatible with oxidizing and reducing substances or strong acids and bases.

13 Method of Manufacture

Alitame may be synthesized by a number of routes.^(2,3) For example, 3-(D-alaninamido)-2,2,4,4-tetramethylthietane is dissolved in water and L-aspartic acid *N*-thiocarboxyanhydride is then added in portions with vigorous stirring, maintaining the pH of 8.5–9.5. The pH is then adjusted to 5.5 and *p*-toluenesulfonic acid

monohydrate is added over a period of one hour. The precipitated crystalline *p*-toluenesulfonate salt is collected by filtration. To obtain alitame from its salt, a mixture of *Amberlite LA-1* (liquid anion exchange resin), dichloromethane, deionized water, and the salt is stirred for one hour, resulting in two clear layers. The aqueous layer is treated with carbon, clarified by filtration, and cooled to crystallize alitame.

Alternatively, tetramethylthietane amine is condensed with an *N*-protected form of D-alanine to give alanyl amide. This is then coupled to a protected analogue of L-aspartic acid to give a crude form of alitame. The crude product is then purified.

14 Safety

Alitame is a relatively new intense sweetening agent used primarily in foods and confectionary. It is generally regarded as a relatively nontoxic and nonirritant material.

Chronic animal studies in mice, rats, and dogs carried out for a minimum of 18 months at concentrations >100 mg/kg per day exhibited no toxic or carcinogenic effects. In people, no evidence of untoward effects were observed following ingestion of 15 mg/kg per day for two weeks.

Following oral administration 7–22% of alitame is unabsorbed and excreted in the feces. The remaining amount is hydrolyzed to aspartic acid and alanine amide. The aspartic acid is metabolized normally and the alanine amide excreted in the urine as a sulfoxide isomer, as the sulfone, or conjugated with glucuronic acid.

The WHO has set an acceptable daily intake of alitame at up to 0.1 mg/kg body-weight.⁽⁴⁾

LD₅₀ (mouse, oral): >5 g/kg

LD₅₀ (rabbit, skin): >2 g/kg

LD₅₀ (rat, oral): >5 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Eye protection and gloves are recommended. Alitame should be stored in tightly closed containers, and protected from exposure to direct sunlight and higher than normal room temperatures.

16 Regulatory Status

Alitame is approved for use in food applications in a number of countries worldwide including Australia, Chile, China, Mexico, and New Zealand.

17 Related Substances

Acesulfame potassium; aspartame; saccharin; saccharin sodium; sodium cyclamate.

18 Comments

The PubChem Compound ID (CID) for alitame is 64763.

19 Specific References

- 1 FAO/WHO. Evaluation of certain food additives and contaminants. Fifty-ninth report of the joint FAO/WHO expert committee on food additives. *World Health Organ Tech Rep Ser* 2002; No. 913.
- 2 Sklavounos C. Process for preparation, isolation and purification of dipeptide sweeteners. United States Patent No. 4,375,430; 1 Mar, 1983.
- 3 Brennan TM, Hendrick ME. Branched amides of L-aspartyl-D-amino acid dipeptides. United States Patent No. 4,411,925; 25 Oct, 1983.
- 4 FAO/WHO. Evaluation of certain food additives and contaminants. Forty-sixth report of the joint FAO/WHO expert committee on food additives. *World Health Organ Tech Rep Ser* 1997; No. 868.

20 General References

- Anonymous. Use of nutritive and nonnutritive sweeteners—position of ADA. *J Am Diet Assoc* 1998; 98: 580–587.
- Hendrick ME. Alitame. Nabors L, Gelardi R, eds. *Alternative Sweeteners*. New York: Marcel Dekker, 1991; 29–38.
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- Renwick AG. The intake of intense sweeteners – an update review. *Food Addit Contam* 2006; 23: 327–338.

21 Author

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22 Date of Revision

5 August 2008.

Almond Oil

1 Nonproprietary Names

BP: Virgin Almond Oil
PhEur: Almond Oil, Virgin
USP-NF: Almond Oil

2 Synonyms

Almond oil, bitter; amygdalae oleum virginale; artificial almond oil; bitter almond oil; expressed almond oil; huile d'amande; oleo de amêndoas; olio di mandorla; sweet almond oil; virgin almond oil.

3 Chemical Name and CAS Registry Number

Almond oil [8007-69-0]

4 Empirical Formula and Molecular Weight

Almond oil consists chiefly of glycerides of oleic acid, with smaller amounts of linoleic and palmitic acids. The PhEur 6.0 describes almond oil as the fatty oil obtained by cold expression from the ripe seeds of *Prunus dulcis* (Miller) DA Webb var. *dulcis* or *Prunus dulcis* (Miller) DA Webb var. *amara* (DC) Buchheim or a mixture of both varieties. A suitable antioxidant may be added.

The USP32–NF27 describes almond oil as the fixed oil obtained by expression from the kernels of varieties of *Prunus dulcis* (Miller) D.A. Webb (formerly known as *Prunus amygdalus* Batsch) (Fam. Rosaceae) except for *Prunus dolcii* (Miller) D.A. Webb var. *amara* (De (Andolle) Focke).

5 Structural Formula

See Section 4.